

Mahaveer Regar

Data Scientist | AI Engineer | ML Engineer

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Education

B.Tech in Chemical Engineering	IIT Hyderabad	2025	CGPA: 7.42
Senior Secondary (XII)	Gyandev Public Sr. Sec. School	2020	98.20%
Secondary (X)	Gyandev Public Sr. Sec. School	2018	93.83%

Professional Experience

Data Scientist | Vipani.ai June 2025 – Present

- Built a performance tracking dashboard for the Catalogue team, enabling daily metric monitoring with delta-based comparisons and batch processing pipelines.
- Designed transformer-based sequence models to predict user web navigation and purchase journeys from clickstream data; achieved 84% next-action prediction accuracy (31% improvement over baseline models) and 76% journey completion prediction accuracy on validation datasets.
- Developed recommendation systems including text-based search, user-item collaborative filtering (ALS), and community-based post recommendation models, improving click-through rate (CTR) from 2.3% to 4.7% and conversion rate from 1.1% to 2.2% through A/B testing in the development environment.
- Collaborated with product and engineering teams to integrate ML models into production workflows.

Research Intern | IIT Hyderabad January 2025 – May 2025

Supervisor: Dr. Lopamudra Giri, Associate Professor

- Built end-to-end machine learning pipeline to classify video-derived data points into distinct clusters, achieving 80% classification accuracy.
- Applied preprocessing including one-hot encoding and sliding window segmentation for LSTM-based time-series modeling.
- Conducted experimental validation of LSTM architectures for temporal sequence classification.

Projects

Agentic Financial Research Assistant | LangGraph, RAG 2025

- Designed and developed a multi-agent financial research assistant using LangGraph, orchestrating Planner, Retriever, Financial Analyst, and Report Writer agents to automate financial document analysis.
- Built a RAG pipeline using ChromaDB for semantic retrieval over annual reports, SEC filings, and earnings transcripts, enabling evidence-grounded analysis and citation-aware report generation.
- Developed an end-to-end application with FastAPI, Streamlit, and Groq LLaMA3, enabling interactive financial research and automated report generation.

Qwen2.5-0.5B Implementation & Financial Fine-Tuning | PEFT 2025

- Recreated the Qwen2.5-0.5B architecture from scratch in PyTorch, implemented core transformer components including RoPE, RMSNorm, Grouped Query Attention (GQA), and KV caching.
- Fine-tuned the model on financial-domain datasets using QLoRA, enabling parameter-efficient adaptation with significantly reduced GPU memory requirements.
- Built an end-to-end training and inference pipeline, demonstrating a deep understanding of modern LLM architecture and efficient fine-tuning techniques.

Technical Skills

Programming: Python, C++, SQL

Machine Learning NLP: Supervised Unsupervised Learning, Ensemble Methods, Recommender Systems, Time Series Forecasting, Text Embeddings

Generative AI: LLMs, Vector Databases, Agentic AI, MCP, LangChain, Fine-tuning (LoRA, QLoRA)

Deep Learning: Neural Networks (ANN, CNN, RNN, LSTM), Attention Mechanisms, Transformers

Frameworks & Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, Pandas, NumPy, Matplotlib, FastAPI

Tools & Platforms: Git, Docker, CI/CD Pipelines (GitHub Actions / GitLab CI), Google Colab